Problem 1:

(a) Pick $a \in G$.

Right identity:

$ax = a$ has a solution $x \in G$. Let, it be e.

$$ae = a$$

Pick arbitrary ~~the~~ $b \in G$ $ya = b$ as solution in G. Let it be y_0.

$$y_0 a = b$$

Now, $(y_0 a)e = be$

$y_0 (ae) = be$ [associativity]

$y_0 a = be$

$b = be$

$\therefore e$ is right identity in G.

Left inverse:

$ya = e$ has a solution in G. Let, it be a^{-1}.

$$a^{-1}a = e$$

So left inverse exist.

Left identity and right inverse:

~~$ea = (a^{-1}a)a$~~

$ae = a(a^{-1}a)$

$a = a(a^{-1}a)$

$$aa^{-1} = (a(a^{-1}a)a^{-1} = (aa^{-1})(aa^{-1})$$

Let $aa^{-1} = d$

$$d = dd$$

d^{-1} exist such that

$$d^{-1}d = (d^{-1}d)d$$
$$e = d$$
$$aa^{-1} = e$$

$\therefore a^{-1}$ is two-sided inverse of a.
$\therefore$ Two-sided inverses exist in G_1.

Then, $a(a^{-1}a) = ea$
$ea = a = ae = e$

$\therefore e$ is also two-sided identity.

We have everything to show that G_1 is a group - associativity, identity and inverse.

$\therefore G_1$ is a group.

(b) Given, $x^2 = e \quad \forall x \in G_1$

That is, $x = x^{-1}$

Let, $x = ab$ for $a, b \in G_1$

Then, $ab = (ab)^{-1} = b^{-1}a^{-1} = ba$

$\therefore G$ is abelian.

(c) ~~Given~~

Proving $\phi(x) = x^{-1}$ is homomorphism $\Rightarrow G$ is abelian.

If $\phi(x) = x^{-1}$ is homomorphism,

i.e. $\phi(xy) = \phi(x)\phi(y)$

$$(xy)^{-1} = x^{-1}y^{-1}$$
$$y^{-1}x^{-1} = x^{-1}y^{-1}$$
$$(y^{-1}x^{-1})^{-1} = (x^{-1}y^{-1})^{-1}$$
$$xy = yx$$

$\therefore G$ is abelian.

Opposite direction:

If G is abelian, $\forall x, y \in G$,

$$\phi(xy) = (xy^{-1}) = y^{-1}x^{-1} = x^{-1}y^{-1} = \phi(x)\phi(y)$$

$\therefore \phi(x) = x^{-1}$ is a homomorphism

Thus, $\phi(x) = x^{-1}$ is a homomorphism $\Leftrightarrow G$ is abelian.

__/__/____

Problem 2 :

Roll number = 2026113005
Then, $d = 5$
Then, $v = 5 \bmod 4 = 1$

$\therefore n = 40$ and $H = \{1, 9\}$

(a) H contains the identity 1

Also, $1^{-1} = 1$
And, $9^{-1} = 9$ $\quad [\because 9^2 = 81 \equiv 1 \bmod 4]$
$\therefore$ inverses exist

~~$1^n \in 1$~~ $\quad 1^n = 1 \in N$
$9^n = 1$ or $9 \in N$
$1^n 9^m = 9^m \in N$
$\therefore$ H is also closed under multiplication.

$\therefore$ H is a subgroup of G.

And, $|H| = 2$

$U(40) = \{1, 3, 7, 9, 11, 13, 17, 19, 21, 23, 27, 29, 31, 33, 37, 39\}$

$\therefore |U(40)| = 16$

$\therefore$ number of cosets $= \frac{16}{2} = 8$ $\quad \left[= \frac{|U(40)|}{|H|}\right]$ [by Lagrange's theorem]

The eight cosets are:

$H = \{1, 9\} = 9H$ $\quad 3H = \{3, 27\} = 27H$

$7H = \{7, 23\} = 23H$ $\quad 11H = \{11, 19\} = 19H$

$13H = \{13, 37\} = 37H$ $\quad 17H = \{17, 33\} = 33H$

$21H = \{21, 29\} = 29H$ $\quad 31H = \{31, 39\} = 39H$

(b) $U(40)$ is an abelian group under multiplication modulo 40. Since, its abelian, H is a normal subgroup. Hence, product of cosets of H is well-defined.

$\therefore U(40)/H$ is a group with identity H.

Cayley table:

	H	$3H$	$7H$	$11H$	$13H$	$17H$	$21H$	$31H$
H	H	$3H$	$7H$	$11H$	$13H$	$17H$	$21H$	$31H$
$3H$	$3H$	H	$21H$	$17H$	$31H$	$11H$	$7H$	$13H$
$7H$	$7H$	$21H$	H	$13H$	$11H$	$31H$	$3H$	$17H$
$11H$	$11H$	$17H$	$13H$	H	$7H$	$3H$	$31H$	$21H$
$13H$	$13H$	$31H$	$11H$	$7H$	H	$21H$	$17H$	$3H$
$17H$	$17H$	$11H$	$31H$	$3H$	$21H$	H	$13H$	$7H$
$21H$	$21H$	$7H$	$3H$	$31H$	$17H$	$13H$	H	$11H$
$31H$	$31H$	$13H$	$17H$	$21H$	$3H$	$7H$	$11H$	H

(c) $U(40)/H$ is isomorphic to $U(10)$

$U(16) = \{1, 3, 5, 7, 9, 11, 13, 15\}$

Define $\phi : U(40)/H \to U(16)$ as:

xH	H	$3H$	$7H$	$11H$	$13H$	$17H$	$21H$	$31H$
$\phi(xH)$	1	3	7	9	15	11	5	13

We can verify that this mapping also preserves the production

Verification examples: Either $(3H)(7H) = 21H$ and image $3 \cdot 7 = 5$
$(7H)(11H) = 13H$ and image $7 \cdot 9 = 15$

Question 3

(a) Let, $x \in G$ has finite order m. Then, $x^m = e$.

Now $\phi(x)^m = \phi(x^m) = \phi(e) = e$

∴ order of $\phi(x)$ is same as that of x

It can be similarly shown in reverse that $\phi(x)^h = e$
$\Rightarrow x^h = e$

(b) $U(16) = \{1, 3, 5, 7, 9, 11, 13, 15\}$
$U(20) = \{1, 3, 7, 9, 11, 13, 17, 19\}$

Define $\phi : U(16) \to U(20)$ as

~~$x \in U(16)$ 1 3 9 11 15 13~~
~~$\phi(x) \in U(20)$~~

$x \in U(16)$	1	3	5	7	9	11	13	15
$\phi(x) \in U(20)$	1	13	3	11	9	17	7	19

It can be verified by making Caley tables for both and seeing that images of products are generated as expected.

Alternatively:

$U(16)$ is generated by 3 and 15 i.e, products of 3 and 15 generate all elements of $U(16)$:

$$1, 3, 3^2=9, 3^3=11, 15, 3\cdot 15=13, 3^2\cdot 15=7, 3^3\cdot 15=5$$

Similarly, $U(20)$ is generated by 13 and 19:

$$1, 13, 13^2=9, 13^3=17, 19, 13\cdot 19=7, 13^2\cdot 19=11, 13^3\cdot 15=3$$

We can clearly see the the mapping

$$\phi(3^a 15^b) = 13^a 19^b \quad \text{is bijective}$$

$\therefore \phi: U(16) \to U(20)$ is a homomorphism.

(c) $U(16)$ has no element of order 8 while $\mathbb{Z}_8$ has one.
$\therefore U(16) \ncong \mathbb{Z}_8$

All elements of $U(24)$ except identity have order 2. While $3 \in U(16)$ has order 4.
$\therefore U(16) \ncong U(24)$

Since, $U(20) \cong U(16)$, $U(20) \ncong \mathbb{Z}_8$ and $U(20) \ncong U(24)$.

□

31/10/26

Problem 4:

(a) Let, $x \in R$, $x \neq 0$.

~~Then, $xr \mapsto$~~

Define $f: R \to R$ ~~$r \mapsto$~~ $r \mapsto xr$

f is injective:

$$xr = xs$$
$$x(r-s) = 0$$
$$r - s = 0 \quad [\because x \neq 0]$$
$$r = s$$

$\therefore$ f is also surjective [injective function from a finite set to itself is surjective]

$\therefore$ $1 = xr$ for some $r \in R$.
$\therefore$ x is a unit.

Now if, $xu = 0$ and $xv = 1$ for $u, v \in R$,

~~$u = 0x = 0v$~~

$$v(xu) = v0$$
$$(vx)u = 0$$
$$u = 0$$

Therefore, a unit can't be a zero-divisor.

31/10/26

(b) We know, $\mathbb{Z}_p[i] \cong \mathbb{Z}_p[x]/\langle x^2+1 \rangle$

Proving $\mathbb{Z}_p[i]$ is a field $\Rightarrow$ -1 is not square in $\mathbb{Z}_p$;

x^2+1 has no factors in $\mathbb{Z}_p[x]$.
$\Rightarrow x^2+1$ has no roots in $\mathbb{Z}_p$

$\therefore \nexists\ k \in \mathbb{Z}_p,\ k^2+1 \equiv 0 \pmod p$

$\Rightarrow k^2 \not\equiv -1 \pmod p$

$\therefore -1$ is not a square in $\mathbb{Z}_p$.

Proving other direction:

$k^2 \not\equiv -1 \pmod p$ for $k \notin \mathbb{Z}_p$
$\Rightarrow k^2+1 \neq 0$
$\Rightarrow x^2+1$ has no roots in $\mathbb{Z}_p$
$\Rightarrow x^2+1$ has no factors in $\mathbb{Z}_p[x]$.
$\Rightarrow \langle x^2+1 \rangle$ is a maximal ideal
$\Rightarrow \mathbb{Z}_p[x]/\langle x^2+1 \rangle$ is a field

$\therefore$ proved

(c) Only zero divisor of $\mathbb{Z}_p[i]$ is $0+0i = 0$.

Now, $(1+i)(1-i) = 1-i^2 = 2$
$(1+i)(1-i)4 = 2 \cdot 4 = 8 = 1 \pmod 7$

$\therefore (1+i)^{-1} = 4(1-i) = 4-4i = 4+3i$ $\quad [-4 = 3 \bmod 7]$

Question 5:

~~Given~~ $\mathbb{Z}_2$

Given, $F = \mathbb{Z}_2[x]/\langle x^2+x+1 \rangle$

(a) Every polynomial has a remainder of degree less than 2 when divided (quotiented) by x^2+x+1

In this case, elements of F are of for $ax+b$ for $a, b \in \mathbb{Z}_2$

$\therefore$ number of elements $= 2^2 = 4$

And, $F = \{0, 1, x, 1+x\}$ ①

Note: In $\mathbb{Z}_2$, subtraction is same as addition:

$-0 = 0 \quad \because 0-0 = 0 \pmod 2$

$-1 = 1 \quad \because 1+1 = 0 \pmod 2$

↓

additive inverses

$\therefore -a = a$

$\therefore a - b = a + (-b) = a + b$

In F, $x^2 + x + 1 = 0$

$\Rightarrow x^2 = x + 1$ [from ①]

Addition table:

+	0	1	x	1+x
0	0	1	x	1+x
1	1	0	1+x	x
x	x	1+x	0	1
1+x	1+x	x	1	0

Multiplication table:

	0	1	x	1+x
0	0	0	0	0
1	0	1	x	1+x
x	0	x	1+x	1
1+x	0	1+x	1	x

(b) Addition is an abelian group:

Addition table shows 0 is additive identity, each element has additive inverse

Commutativity is visible from symmetry.

Associativity is inherited from polynomial add^{n}

Multiplication:

Associativity is inherited.

Commutativity and identity are visible in multiplication table.

And every non-zero element has an inverse (from table).

Distributivity: Inherited from polynomial arithmetic.

Therefore, F is a field.

(b) F group under multiplication, $F^\times = \{1, x, 1+x\}$

Now observe,

$x' = x$

$x^2 = 1+x$

$x^3 = x(1+x) = x+x^2 = x+(1+x) = 1$

$\therefore$ powers repeat as: $1, x, 1+x, 1, x, 1+x, \ldots$

(↓ x^0)

And they include all non-zero elements.

Therefore, $F^\times = \langle x \rangle$

(c) Smallest $y \in F$ st $1+y=0$ is $y=1$.

$\because 1+1 = 2 \ (\equiv 0)$ is characteristic of F by definition.

Given, $\phi: F \to F$ st $\phi(a) = a^2$

Now, $\phi(a+b) = (a+b)^2 = a^2+b^2+2ab = a^2+b^2$
$= \phi(a^2) + \phi(b)$

$2ab \equiv 0 \pmod 2$

And, $\phi(ab) = (ab)^2 = a^2b^2 = \phi(a)\,\phi(b)$.

Therefore, ϕ preserves both operations and also $1 \mapsto 1$.

Seeing the isomorphism

$\phi(0) = 0 \quad \phi(1) = 1 \quad \phi(x) = 1+x \quad \phi(\overset{1+x}{\cancel{x^2}}) = x$

These values are distinct.

Therefore, ϕ is a ring isomorphism from F to itself.

31/10/26

Question 5:

(a) Let, $\frac{\alpha}{\beta} = x + iy \qquad x, y \in \mathbb{R}$

Choose $m, n \in \mathbb{Z}$ such that,

$$|x-m| \leq \frac{1}{2} \text{ and } |y-n| \leq \frac{1}{2}. \qquad -①$$

Now set $q := m + in$

$$\begin{aligned} r &:= \alpha - q\beta \\ &= \beta\left(\frac{\alpha}{\beta} - q\right) \\ &= \beta\left((x-m) + (y-n)i\right) \end{aligned}$$

Now,

$$\begin{aligned} N(r) &= N(\beta)\, N\left((x-m) + (y-n)i\right) \\ &= N(\beta)\left((x-m)^2 + (y-n)^2\right) \end{aligned}$$

From ①,

$$(x-m)^2 \leq \frac{1}{4}$$
$$(y-n)^2 \leq \frac{1}{4}$$
$$\Rightarrow (x-m)^2 + (y-n)^2 \leq \frac{1}{2}$$

$$\therefore N(r) \leq \frac{1}{2} N(\beta) < N(\beta)$$
$$\therefore N(r) < N(\beta)$$

31/10/26

(b) ~~Let, $I = \langle 0 \rangle$ is principal ideal of $\mathbb{Z}[i]$~~

Let, I is an ideal of $\mathbb{Z}[i]$.

Case I: $I = \langle 0 \rangle$

It is trivially principal.

Case II: If Otherwise:

Let, $\beta \neq 0$, $\beta \in I$ such that $N(\beta)$ is as small as possible.

Using (a), for any $\alpha \in I$, $\alpha = q\beta + r$ $q, r \in \mathbb{Z}[i]$
s.t, $N(r) < N(\beta)$.

Since I is an ideal, $r = \alpha - q\beta \in I$.

~~Since $N(\beta)$~~ Choice of β forces $r = 0$. [if $r \neq 0$, $N(r) < N(\beta)$. This contradicts our choice]

$\therefore \alpha = q\beta$
$\Rightarrow \alpha \in I$
$\Rightarrow I \subseteq \langle \beta \rangle$

Also, $\beta \in I \Rightarrow \langle \beta \rangle \subseteq I$

$\therefore \langle \beta \rangle = I$

$\therefore$ every ideal of $\mathbb{Z}[i]$ is principal. $\therefore$

31/10/26

(c) $5 = 4+1 = 4 - i^2 = (2+i)(2-i)$

$3+i = 3(2+1) + (2-1)i = (2-i) + i(2-i)$
$= (2-i)(1+i)$

∴ multiples of 5 and $3+i$ and their products are all multiples of $2-i$.

∴ $\langle 5, 3+i \rangle \subseteq \langle 2-i \rangle$.

And, $5 - (3+i) = 2-i \in \langle 3+i, 5 \rangle$
$\Rightarrow \langle 2-i \rangle \subseteq \langle 3i+5 \rangle$

Therefore, $\langle 5, 3+i \rangle = \langle 2-i \rangle$